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SCARF-SLAM: Spatially Confidence-aware Aggregation for Federated Implicit Neural SLAM

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***Abstract*—Rapid and accurate environmental reconstruction is a foundation for intelligent mobility systems, such as autonomous vehicles, mobile robots, and drones, to navigate and understand complex spaces. Neural implicit representations offer an effective way to model continuous 3D scenes for dense visual SLAM, but most existing methods depend on centralized training with a single agent, which limits their scalability in large environments. In realistic scenarios, multiple agents explore different parts of a scene, resulting in highly unbalanced observations that make model fusion non-trivial. In this paper, we propose SCARF-SLAM, a federated implicit neural SLAM framework that enables collaborative large-scale mapping through spatially confidence-aware aggregation. SCARF-SLAM leverages confidence derived from volume rendering to aggregate spatial implicit features across agents, while regulating the fusion of global decoder parameters according to their effective exploration. This unified aggregation strategy allows robust learning under heterogeneous data distributions without sharing raw sensor data. Experiments on large-scale datasets show that SCARF-SLAM achieves accurate reconstruction and stable SLAM performance, outperforming the existing NeRF-based methods and naive federated baselines.**

***Index Terms*—Neural SLAM, Federated Learning, Implicit Scene Representation, Confidence-aware Aggregation**

I. Introduction

Reliable environmental perception is a prerequisite for the safe and efficient operation of intelligent mobility systems, including autonomous vehicles [1], mobile robots [2], and augmented reality applications [3]. In this context, reconstructing 3D scene environment offers both the estimated poses of a camera and a dense map of the surrounding environment in real-time, playing a vital role in these fields. Traditional methods for 3D reconstruction generally rely on feature tracking [4] [5] or direct matching algorithms [6] [7] in the frontend to get camera poses and 3D coordinates in optimization backend. Although traditional methods achieve successful results, they face challenges in error accumulation, limited adaptability, poor expansion in unobserved regions, and high computational and memory costs.

Recently, Neural Radiance Field (NeRF) [8] has been proposed to reconstruct the 3D surface of the environment. Utilizing Multi-Layer Perceptions(MLPs) to model the surface of the geometry scene, it provides a compact yet powerful representation which is capable of rendering photorealistic scenes with minimal storage requirements. iMAP [9] and NICE-SLAM [10], as the pioneering two NeRF-based SLAM systems, proves their ability to offer a detailed and continuous surfaces reconstruction.

An obvious advantage of this Neural Implicit Representations (NIRs) lies in their ability to capture essential information in high-dimension data structure with fewer parameters. Furthermore, the neural networks used in NIRs enable encoding continuous and differentiable rendering, facilitating more efficient integration with deep learning pipelines through gradient-based optimization techniques [10]. Since the NIR method models the space as a continuous and differentiable function, it also build the vision foundation for related works such as predicting wireless channels [11] and reconstructing the radio map [12] [13]. When the channel state or radio environment changes, just a few parameters need to be adjusted and the implicit function will learn itself to adapt the changes with the gradient descent algorithm.

Although neural implicit representations enable expressive and continuous scene modeling, their application to SLAM introduces substantial computational overhead as the mapped environment grows. The increasing cost of optimization and rendering makes it difficult for a single-agent system to sustain real-time performance over large scenes [10]. A natural way to mitigate this limitation is to leverage multiple agents that operate concurrently, each contributing partial observations of the environment [14]. By distributing the mapping process across several platforms, the overall computational load can be effectively reduced while improving coverage efficiency. Such a collaborative setting closely reflects practical deployment scenarios, for example teams of micro aerial vehicles jointly

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Fig. 1: Qualitative comparison of our proposed SCARF-SLAM with existing NeRF-based SLAM systems, iMAP* [9] and NICE-SLAM [10] on scene0000 of the ScanNet dataset [17]. Our method produces more accurate geometric structure with higher-quality texture.

exploring an environment [15], and provides a scalable path toward building richer and more complete scene reconstructions [16].

To support efficient collaboration among multiple agents, we adopt a federated learning paradigm for neural implicit SLAM. In this setting, each agent optimizes its own implicit model using locally observed data, while a central server coordinates model fusion without accessing raw sensor measurements. This design not only reduces centralized computation but also avoids direct data sharing between agents. Nevertheless, directly applying existing federated learning strategies to neural implicit representations is challenging. The highly coupled parameters of implicit models make naive model averaging unreliable, and the uneven spatial coverage produced by different exploration trajectories introduces severe heterogeneity across local models, which can significantly hinder stable convergence.

To address the aforementioned challenges, we introduce **SCARF-SLAM** (Spatially Confidence-aware AggRegation for Federated Implicit Neural SLAM), a federated neural SLAM framework designed for scalable and accurate 3D scene reconstruction. SCARF-SLAM adopts a tri-plane implicit representation [18] that supports efficient local optimization on individual agents while remaining compatible with federated model fusion. Instead of relying on uniform parameter averaging, SCARF-SLAM explicitly models the reliability of learned spatial features under heterogeneous exploration. By accumulating confidence derived from the rendering process, the framework assigns spatially varying importance to feature planes during aggregation, effectively mitigating artifacts caused by sparsely observed regions. In addition, SCARF-SLAM incorporates proximal regularization during local optimization to stabilize federated training, reducing update divergence and promoting consistent convergence across communication rounds.The main contributions of this work are summarized as follows:

- We present **SCARF-SLAM**, a federated neural implicit SLAM framework that enables collaborative mapping in large-scale environments.
- We propose a spatially confidence-aware aggregation strategy that accounts for heterogeneous exploration and improves the fusion of implicit scene representations across agents.
- We integrate proximal regularization into local neural implicit optimization to enhance the stability and convergence of federated SLAM training.

II. Related Works

A. Dense Visual SLAM

Visual SLAM systems produce two types of maps in terms of reconstructing the scene: sparse and dense. Sparse SLAM systems [19] [5] [4] mainly target at recovering the accurate camera poses and reconstructing the coarse scene map using feature points. Although sparse SLAM systems can achieve real-time performances, in most domains such as robotics, virtual/augmented reality, a globally consistent dense map is often needed. Dense SLAM systems [20] [21] [22] simultaneously estimate the camera trajectory and reconstruct the dense geometry or surface of the environment. They often represent the geometric scene as volume [20] [23] or surfels [21] [24]. While these SLAM systems can provide dense maps for high-precision applications like navigation, they often struggle to find a balance between resolution and memory consumption.

With the emergence of deep learning, some methods [25] [26] [27] [28] [29] use this data-driven prior to tackle the vulnerability to textureless regions and illumination changes existing in traditional visual slam. Early works like PoseNet [30] and DeepVO [31] use convolutional neural networks (CNNs) or recurrent neural networks (RNNs) to predict the camera poses in end-to-end regression, but they lack of geometric constrains and generalization. Some contemporary SLAM systems like DROID SLAM [25] turn to a hybrid architecture, combining the deep feature extraction with differentiable bundle adjustment, and achieve amazing results.

B. NeRF-based SLAM

Recently NeRF [8] has been proven to be a promising method in various 3D vision tasks such as novel view synthesis [32] [33], object-level reconstruction [34] [35] and scene-level reconstruction [36] [37] [38]. However, it requires hours of training to converge. Several NeRF variants [33] [39] [40] focus on improving scalability and rendering quality, while others try to accelerate training process by leveraging diverse representation methods such as points [41], voxels [42] [43] and hash table [44]. Based on these techniques, some NeRF-based SLAM systems have been proposed, demonstrating theirs abilities at generating high quality dense maps. iMAP [9] is the first SLAM systems to combine NeRF into parallel mapping and tracking threads. NICE SLAM [10] uses hierarchical MLPs rather than a single MLP to encode the entire scenes, attempting to expand to large-scale reconstruction. NICER SLAM [45] and NeRF-SLAM [46] build a unified SLAM system with a only monocular RGB sequence. Point-SLAM [47], Vox-Fusion [48] and ESLAM [49] leverage various representation methods in point cloud, voxels and tri-plane encoding for more precise reconstruction.

Despite the above SLAM systems perform excellent reconstruction results, all of them use just a single agent to explore scenes, which actually hinder the reconstructing and converging speed of the whole map. Our proposed approach SCARF-SLAM employs multiple agents to work collaboratively and guarantees real-time and high fidelity reconstruction.

Recently, 3D Gaussian Splatting(3DGS) [50] has emerged as a popular alternative to NIRs, offering rapid rendering speeds and explicit geometric manipulation. While 3DGS excels in rendering speed, our work specifically focuses on neural implicit representations (SDFs) due to their continuous, memory-efficient nature and smooth gradient properties. These characteristics are particularly advantageous for federated aggregation across devices with limited communication bandwidth, providing a different trade-off between memory footprint and rendering speed.

C. Federated Learning on Local Implicit Models

Federated learning enables multiple agents to collaboratively optimize a shared model while keeping all sensor data local to each participant. Each agent updates model parameters using its own observations, and a central server periodically combines these updates to form a global solution. This optimization process is commonly expressed as

$$\min_{w} \sum_{k=1}^{K} p_k F_k(w), \tag{1}$$

where $F_k(w)$ denotes the local objective of agent k and p_k controls its contribution during aggregation.

While FedAvg [51] is a powerful averaging approach under homogeneous conditions, it becomes unreliable when local training dynamics or data distributions differ across agents. To improve robustness, proximal regularization methods like FedProx [52] constrain local updates to remain close to the global model, reducing divergence during federated optimization. These challenges are further amplified for neural implicit representations, whose parameters are tightly coupled and highly sensitive to uneven spatial supervision.

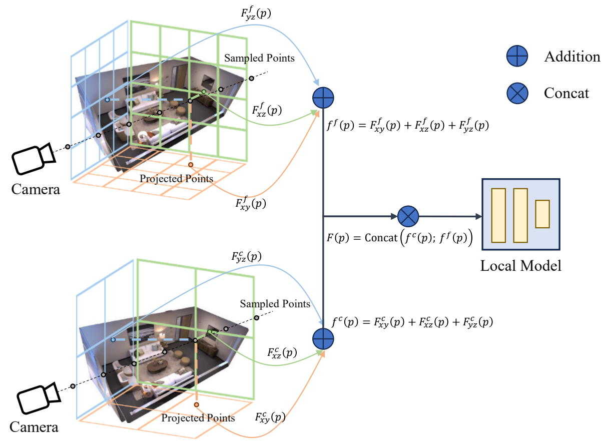

Fig. 2: The local implicit representation of our method. At an estimated camera pose$\{R, t\}$, we cast a ray for each pixel and sample total N points$\{p_n\}_{n=1}^{N}$, where each point p_n is projected on three coarse planes and three fine planes. The features are concatenated together into the two-layers MLPs decoder to output the raw SDF values and color.

Several works [53] [54] [55] have achieved scalable scene modeling by decomposing environments into localized neural fields, yet they typically depend on shared raw data and offline training. Meanwhile, existing multi-agent SLAM systems [16] [56] support distributed mapping but heavily rely on explicit coordination—such as cross-agent loop closures—to enforce global consistency. In contrast, our method introduces a federated fusion strategy for neural implicit representations. Through this framework, each agent naturally absorbs global spatial context from the shared model, drastically reducing the reliance on continuous loop detection during active exploration.

III. Problem Formulation

Suppose there are N agents, labeled as $C_1, \cdots, C_N$, exploring a set of M scenes, denoted by $S_1, \cdots, S_M$. Agent C_i explores the scene S_m^i and subsequently trains the representation $\{\mathcal{V}_i^{local}, \mathcal{W}_i^{local}\}$. Our goal is to obtain a comprehensive global scene representation $\{\mathcal{V}^{global}, \mathcal{W}^{global}\}$ that covers the entire set of scenes $\mathcal{S} \triangleq \bigcup_{i \in M} S_m$ with the assistance of a central server, all while avoiding the exchange of raw data. Given that the trajectories of agents are inherently random during exploration, the aim of FL is to minimize the loss function defined as:

$$\arg\min_{\mathcal{V},\mathcal{W}} \mathcal{L}(\mathcal{V}, \mathcal{W}) = \sum_{i}^{M} \frac{|S^i|}{|\mathcal{S}|} L_i(\mathcal{V}, \mathcal{W}) \tag{2}$$

where $L_i(\mathcal{V}, \mathcal{W})$ is the presentation loss on the scene S_i, and $|S^i|$ denotes the geometry size of scene S_i.

To address the challenge of unbalanced data distributions resulting from the random exploration patterns of multiple agents, and leads to the corresponding heterogeneous decoder, we introduce a specialized confidence-aware FL approach for effective aggregation of local NIR models.

IV. Our Method

Our approach centers on building a FL framework to aggregate the Signed Distance Field (SDF) network model collected and training from multiple agents while minimizing the necessity for extensive data transmission. Achieving this objective hinges on two key components. Firstly, we leverage a tri-plane scene representation that can not only satisfy large-scale scenes but is also seamlessly integrated into a FL pipeline (Sec. IV-A). Secondly, we derive a cross-silo FL scheme aiming to overcome the unique challenges associated with NIR models on unbalanced data distributions (Sec. IV-C).

A. Local Implicit representation

In this part, we introduce our local scene implicit representation model. Since the voxel grid-based methods suffer from cubic memory growth issue, inspired by ESLAM [49], we employ a tri-plane architecture in which we store and optimize features on perpendicular axis-aligned planes. Specifically, we use two-scale tri-plane representation, in coarse and fine, to encode the scene(as illustrated in Fig. 2). The tri-plane features $\boldsymbol{F}(p)$ can be expressed as:

$$\begin{aligned} f^c(p) &= F^c_{xy}(p) + F^c_{xz}(p) + F^c_{yz}(p) \\ f^f(p) &= F^f_{xy}(p) + F^f_{xz}(p) + F^f_{yz}(p) \\ \boldsymbol{F}(p) &= \text{Concat}(f^c(p); f^f(p)) \end{aligned} \tag{3}$$

where $f^c(p)$ and $f^f(p)$ denote the feature planes in coarse and fine resolutions. p is the world coordinate. $\{F^c_{xy}(p), F^c_{yz}(p), F^c_{xz}(p)\}$ are the three coarse feature planes and $\{F^f_{xy}(p), F^f_{yz}(p), F^f_{zx}(p)\}$ are the fine feature planes. For a sample point p in a ray cast by a camera, we use bilinear interpolating the four nearest neighbors on each feature plane. Then we sum the interpolated coarse features and fine features into the coarse output $f^c(p)$ and the fine output $f^f(p)$ respectively. Lastly, the coarse and fine features are concatenated together into a two-layer MLPs to output the predicted SDF values and color. Using tri-plane method to extract features allows us to achieve more accurate and real-time reconstruction with less memory usage.

B. Individual training with Dense SLAM

For the training of local implicit agents, we follow the practice of NICE SLAM [10] and select a batch of rays R with ground truth depths. In online SLAM, achieving close-to-frame-rate camera tracking is crucial for the robust optimization of smaller displacements. Therefore, a parallel tracking and mapping process [9] is employed, which continuously optimizes the pose (rotation R and translation t) of the latest

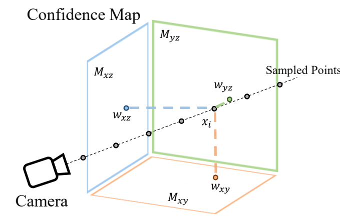

Fig. 3: The confidence-aware feature aggregation in our method. We project the rendering weights w_i of a sample pointx_i onto the corresponding axis-aligned planes to form the confidence map $\mathcal{M} = \{M_{xy}, M_{yz}, M_{xz}\}$. The confidence map $\mathcal{M}$ helps us to assess how much the sample points contribute to the surfaces reconstruction.

selected keyframes and combines the loss of mapping and tracking. The depth and color rendering loss of a pixel as:

$$\mathcal{L}_c = \frac{1}{|R|}\sum_{r\in R}(I(r) - \hat{I}(r))^2, \quad \mathcal{L}_d = \frac{1}{|R|}\sum_{i\in R}(D(r) - \hat{D}(r))^2 \tag{4}$$

where $I(r)$ and $D(r)$ are the color and depth of the pixel of ray r and $\hat{I}(r)$ and $\hat{D}(r)$ are the rendered color and depth based on the surface reconstruction. Additionally, we introduce the SDF loss and free space loss for our mapping thread. Specifically, for pixel samples in truncation region, we use the depth sensor measurement to estimate the signed distance field:

$$\mathcal{L}_T(P^T_r) = \frac{1}{|R|}\sum_{r\in R}\frac{1}{|P^T_r|}\sum_{p\in P^T_r}(z(p) + \phi_{\boldsymbol{g}}(p)\cdot T - D(r))^2 \tag{5}$$

where P^T_r is a set of points on ray r that lie in truncation region, $z(p)$ is the planar depth of the point p to the camera, and $|z(p) - D(r)| < T$. We differentiate the importance of points that closer to the surface $P^{T-m}_r = \{p|z(p) - D(r) < 0.4T\}$ and define the points at the tail of the region $P^{T-t}_r = P^T_r - P^{T-m}_r$, then:

$$\mathcal{L}_{T-m} = \mathcal{L}_T(P^{T-m}_r) \quad \text{and} \quad \mathcal{L}_{T-t} = \mathcal{L}_T(P^{T-t}_r) \tag{6}$$

For sample points out of the truncation region, we define the SDF loss as:

$$\mathcal{L}_{fs} = \frac{1}{|R|}\sum_{r\in R}\frac{1}{|P^{fs}_r|}\sum_{p\in P^{fs}_r}(\phi_{\boldsymbol{g}}(p) - 1)^2 \tag{7}$$

This allows us to limit the SDF value within the truncation region. Through gradient descent optimization, we jointly optimize the feature planes, MLP decoder and camera poses.

C. Confidence-aware Feature Aggregation

In this part, we propose our confidence-aware feature aggregation approach. While the coordinate-based updates in local implicit agents align well with FedAvg [51], the diverse trajectories of agents lead to unevenly explored feature volumes.

Method	Reconstruction (cm)				Localization (cm)	
	Depth L1↓	Acc.↓	Comp.↓	Comp. Ratio (%)↑	ATE Mean↓	ATE RMSE↓
iMAP* [59]	11.45	6.68	7.69	49.00	7.75	6.43
NICE-SLAM [87]	4.51	3.36	2.76	90.47	3.43	2.17
SCARF-SLAM (ours)	**1.61**	**1.33**	**1.57**	**96.92**	**0.41**	**0.35**

TABLE I. Quantitative comparison results of our method with existing NeRF-based SLAM systems, iMAP* [9] and NICE-SLAM [10] on Replica [57] dataset. The results are the average of the eight scenes. Our method outperforms previous works by a high margin.The evaluation metrics for reconstruction are L1 loss (cm) between rendered and ground truth depth maps of 1000 random camera poses, reconstruction accuracy (cm), reconstruction completion (cm), and completion ratio (%). The evaluation metrics for localization are mean and RMSE of ATE (cm) [58].

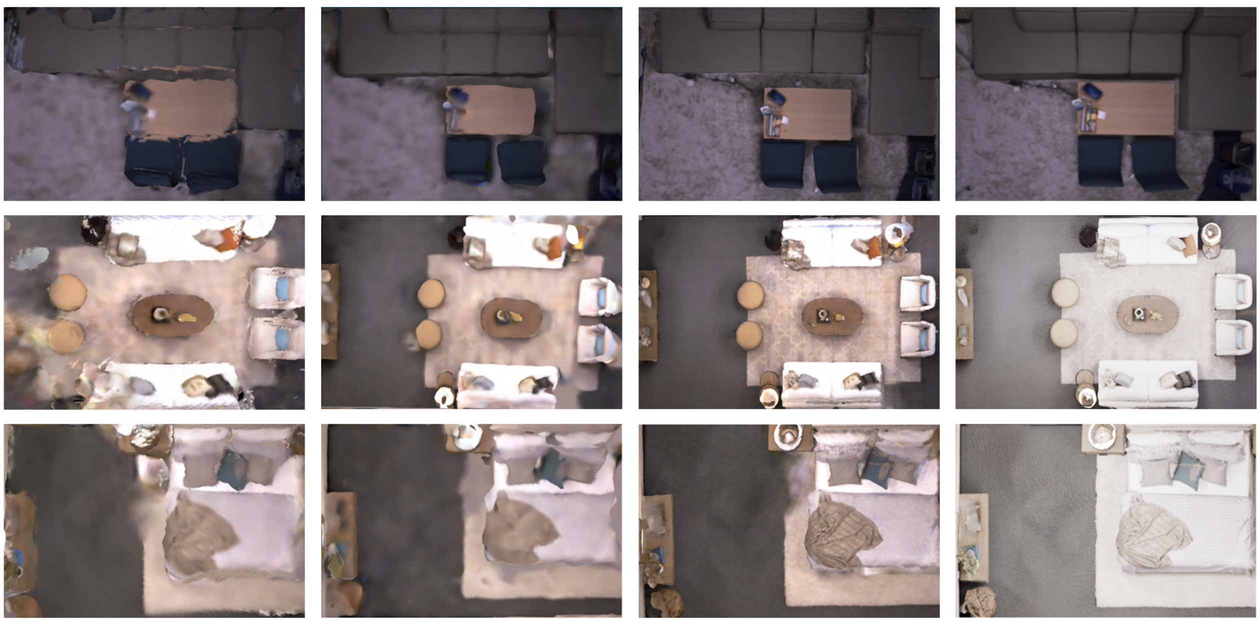

Fig. 4. Qualitative comparison results of our method with existing NeRF-based SLAM systems, iMAP* [9] and NICE-SLAM [10] on the Replica [57] dataset. Our method produces more accurate detailed geometry as well as higher-quality textures.

Specifically, unobserved regions remain initialized near zero, while well-explored areas contain meaningful features. Naive averaging treats these equally, causing un-updated parameters to dilute the global model and introducing artifacts from transient noise. To address these problems, we introduce a *Weight Accumulation* mechanism that leverages the volume rendering process to quantify feature reliability. In SDF-based volume rendering [35], the contribution of a sample point $\mathbf{x}_i$ along a ray $\mathbf{r}$ is determined by its visibility weight $w_i = T_i \alpha_i$, derived from the SDF-based opacity α_i and transmittance T_i. We reintroduce this rendering weight w_i as a proxy for geometric confidence. Specifically, we maintain a set of accumulation maps $\mathcal{M} = \{M_{xy}, M_{yz}, M_{xz}\}$ corresponding to the tri-plane features(see Fig. 3). During ray marching, for a sample $\mathbf{x}_i$ with weight w_i, we project its coordinate onto each plane S and accumulate the weights online:

$$W_S(\mathbf{u}) \leftarrow W_S(\mathbf{u}) + \sum_{\mathbf{r} \in R} \sum_{i=1}^{N} w_i \cdot \delta(\text{proj}_S(\mathbf{x}_i) - \mathbf{u}) \tag{8}$$

where $\delta(\cdot)$ handles grid discretization via scatter-add operations. High accumulated weights $W_S(\mathbf{u})$ indicate surfaces consistently verified by photometric loss, effectively distinguishing valid geometry from empty space or floaters during federated aggregation.

D. Exploration-aware Decoder Aggregation

Unlike the spatially-sparse feature planes, the Multi-Layer Perceptrons (MLPs) used as decoders serve as a global implicit function shared across the entire scene. Since the decoder parameters are location-invariant, a coordinate-based accumulation is not applicable. However, performing a naive average (FedAvg) on the decoder parameters is suboptimal, as agents

Method	ATE	Sc. 0000	Sc. 0059	Sc. 0106	Sc. 0169	Sc. 0181	Sc. 0207	Ave.
iMAP* [9]	Mean	24.99	19.91	14.28	16.64	25.02	11.14	18.66
	RMSE	19.35	16.81	12.80	13.81	20.97	10.25	15.67
NICE-SLAM [10]	Mean	13.88	12.58	8.16	9.8	13.21	5.73	10.56
	RMSE	11.27	10.58	7.23	7.94	11.91	4.97	8.98
SCARF-SLAM (ours)	Mean	**4.67**	**10.83**	**4.53**	**6.48**	**7.69**	**5.50**	**6.62**
	RMSE	**4.2**	**10.04**	**4.05**	**5.85**	**7.07**	**4.73**	**5.99**

TABLE II. Quantitative comparison of ATE (Mean[cm] and RMSE[cm]) [58] for the localization on the ScanNet dataset [17]. Our proposed method outperforms the previous methods by a large margin.

Fig. 5. Qualitative comparison results of our method with existing NeRF-based SLAM systems, iMAP* [9] and NICE-SLAM [10] on the ScanNet [17] dataset.Our method produces more accurate detailed geometry as well as higher-quality textures.

operating in limited or texture-less regions may produce underfitted decoder parameters, which can dilute the generalization capability of the global model.

To address this issue, we propose an exploration-aware weighted aggregation strategy. We assign a scalar confidence score s_k to the k-th agent, proportional to the volume of its validly mapped region (i.e., the number of active voxels in the feature planes). The global decoder parameters θ_{global} are updated by a weighted sum of local parameters θ_k:

$$\theta_{global} = \sum_{k=1}^{K} \frac{s_k}{\sum_{j=1}^{K} s_j} \theta_k, \tag{9}$$

where K is the total number of agents. By normalizing the confidence scores, this strategy ensures that the global decoder is primarily driven by agents with extensive exploration and rich geometric observations, while suppressing the noise from agents with insufficient data.

E. Joint FL Optimization with FedProx

Once the feature and decoder aggregations are finished in the first round, we obtain an accurate initial global model $\{\mathcal{V}_1^{global}, \mathcal{W}_1^{global}\}$. However, simply using FedAvg [51] could lead to an accuracy degradation in the subsequent generated model because no supervision is applied in FL optimization. Therefore, after the first round aggregation, we employ FedProx [52] on each agent in later communication rounds by adding a proximal term $\frac{\mu}{2}|w - w_{global}|^2$ on the loss function during the mapping thread. In this way the local NIR models obtained from the agents are forced to keep close to the global model, thus maintaining more precise for the optimization later.

V. Experiments

In this section, we conduct a comprehensive evaluation on our method and compare it with other NeRF-based SLAM systems and FL algorithms.

A. Experiment setup

Datasets. We consider two diverse datasets, namely Replica [57] and ScanNet [17] dataset and select the same scenes in NICE-SLAM [10] for evaluation.

Benchmarks. We compare our method with several existing state-of-the-art NIR methods. Since iMAP is not open-source, we choose iMAP* which is the re-implementation of iMAP in [10]. In additional, we extend our comparison with FedAvg [51] to demonstrate the effectiveness of our FL strategy.

	ATE Mean	ATE RMSE
FedAvg [51]	12.35	10.59
SCARF-SLAM(ours)	**4.67**	**4.2**

TABLE III. Quantitative comparison of our method's localization accuracy with FedAvg [51] on scene0000 of the ScanNet dataset [17].

Metrics. We conduct an evaluation on both 2D and 3D metrics to access the quality of scene reconstruction following the step of NICE-SLAM [10]. For 2D metrics,we use L1 loss to quantify differences and reconstruction quality between corresponding elements of depth and RGB maps. For 3D metrics, we consider reconstruction accuracy [cm], reconstruction completion [cm] and completion ratio [< 5 cm %]. For evaluating these metrics, we build a TSDF volume for a scene with a resolution of 1 cm and use the marching cubes algorithm [59] to obtain scene meshes. Before evaluating the 3D metrics for our method and for the baselines, we perform mesh culling as recommended in [36]. We use ATE [58] to evaluate camera localization.

Implementation. We run SCARF-SLAM on a NVIDIA GeForce RTX 4090 GPU. For multiple agents experiments, we employ additional NVIDIA RTX 4090 GPUs and set the original one as the central server. We set the truncation distance in 6 cm, the coarse planes resolution in 24 cm and the fine planes resolution in 6 cm. All feature planes have 32 channels, resulting in a 64-channel concatenated feature input for the decoders. The decoder are two-layers MLPs with 32 channels in the hidden layer. The weight μ of the proximal term is set to 10 and the communication round is set to 5. To simulate multi-agent exploration realistically, we divide the dataset sequences into multiple non-overlapping trajectory segments, assigning each segment to an individual agent. The training strategy of our FL algorithm is summarized in Algorithm 1.

Algorithm 1: The SCARF-SLAM framework

Input: N is the number of agents, T is the total communication rounds, fed_freq is the local update frequency.
Result: Global federated model $\{\mathcal{V}_{global}, \mathcal{W}_{global}\}$.
Server Executes:
    initialize the global model $(\mathcal{V}_0, \mathcal{W}_0)$ at $t = 0$;
    for $t = 0, 1, \cdots, T-1$ **do**
        for $n \leftarrow 1, 2, \ldots, N$ ***in parallel*** **do**
            $\{\mathcal{V}^n_{t+1}, \mathcal{W}^n_{t+1}\} \leftarrow \mathrm{AgentTraining}(n, \mathcal{V}_t, \mathcal{W}_t)$;
        $\mathcal{V}_{t+1} \leftarrow \mathrm{FeatureAggregate}(\{\mathcal{V}^n_{t+1}\}^N_{n=1})$;
        $\mathcal{W}_{t+1} \leftarrow \mathrm{DecoderAggregate}(\{\mathcal{W}^n_{t+1}\}^N_{n=1})$;
Return: $\{\mathcal{V}_T, \mathcal{W}_T\}$;
Function *AgentTraining*$(i, \mathcal{V}_{local}, \mathcal{W}_{local})$**:**
    $count\_frames \leftarrow 0$;
    while $count\_frames < fed\_freq$ **do**
        $\mathcal{V}_{local}, \mathcal{W}_{local} \leftarrow \mathrm{SLAM\_Optimize}(\mathcal{V}_{local}, \mathcal{W}_{local})$;
        $count\_frames \leftarrow count\_frames + 1$;
    return $\{\mathcal{V}_{local}, \mathcal{W}_{local}\}$;

B. Experimental Results

Evaluation on Replica [57]. We provide the quantitative analysis of our experimental results on the eight scenes of the Replica dataset [57] in Tab. I. The numbers represent the average of the metrics. As shown in Tab. I, our proposed method outperforms the baselines for both reconstruction and localization accuracy. Qualitative analysis on the Replica dataset [57] is provided in Fig. 4. The results show that our method achieve more accurate reconstruction and produce less artifact and floaters.

Evaluation on ScanNet [17]. We also benchmark ours and existing methods on multiple large scenes from the ScanNet dataset [17] to evaluate and compare their scalability. For evaluating camera localization, we report the average of the mean and RMSE of ATE [58] on six diverse scenes in Tab. II. As demonstrated in the table, our method's localization is more accurate than existing methods.

Since the ground truth meshes of the ScanNet dataset [17] are incomplete, we only provide qualitative analysis for geometry reconstruction, similar to previous works. The qualitative comparisons in Fig. 1 and Fig. 5 validate that our model reconstructs more precise geometry and detailed textures than existing approaches.

Model Aggregation accuracy. To further compare our FL strategy with the FedAvg [51], we split the scene0000 of the ScanNet dataset [17] into 3 parts and evaluate the ATE (Mean[cm] and RMSE[cm]) [58] results. Tab. III demonstrates the enhanced 3D reconstruction performance after model aggregating with our proposed FL strategy.

Computation complexity & Runtime. FL framework can reduce $N\times$ computational time of mapping when there are N agents involved. The aggregation deal with the aggregation of models from multiple agents every iteration with a complexity of $O(Nlog(N))$. Experiments show that our method achieves real-time reconstruction at 6 frames per second.

VI. Conclusion

In this work, we addressed the scalability limits of centralized SLAM by proposing SCARF-SLAM. Through a novel weighted accumulation scheme, SCARF-SLAM successfully bridges the gap between independent local exploration and coherent global reconstruction. Our framework allows heterogeneous agents, such as drones and ground robots, to fuse their spatial knowledge effectively, ensuring accurate mapping even under highly disparate viewing angles. By combining high-fidelity reconstruction with data privacy, SCARF-SLAM provides a viable framework for multi-agent systems to operate in large-scale environments.

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